

AIRAN LIANG - Graphics Programmer

Contact Information

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Summary

Experienced rendering programmer with over 10 years of hands-on expertise in real-time and offline computer graphics. Specialized in implementing advanced rendering features, optimizing rendering pipelines, and bringing cutting-edge graphics research into production. Passionate about graphics technology, actively contributing to open-source projects and continually improving through self-learning. Comfortable working in both AAA studios and smaller teams, and always ready to collaborate and share knowledge.

Core Skills

- **Rendering & Graphics:** Real-time Rendering, Offline Rendering, Computer Graphics, Math for Graphics
- **Tech Stacks:** C/C++, C#, HLSL/GLSL, DirectX, Vulkan.
- **Engines:** Unity, OGRE, Custom In-house Engines.
- **Graphics Tools:** Renderdoc, GPA, NSight, Snapdragon and PIX.
- **Specialties:** GPU-driven Rendering, Material Systems, Shader Optimization, Research Paper Implementation

Personal Experience

Ubisoft Shanghai — *Senior Rendering Programmer*

Dec 2023 – Present | Rainbow Six Siege

- Collaborate with the core rendering team in Montreal to develop, maintain and optimize advanced rendering features for a live AAA title

- Integrated Nvidia DLSS Frame Generation into the rendering pipeline
- Designed and implemented custom VFX rendering systems, including space-based particle effects, particle trails, and optimized decal rendering
- In-engine debug tools (e.g., Quad Overdraw, UV Overlap Check, PIX Debug integration)

Lilith Games (Party Project) — *Senior Rendering Engineer*

Mar 2021 – Oct 2023

- Built and optimized a Unity-based rendering pipeline for a stylized large-scale project
- Reduced shadow caster drawcalls via custom CSM scrolling and Unity source code modification
- Developed a data-driven static batching system
- Implemented GPU-driven instancing via HISM
- Contributed to in-house engine development including texture compression systems and material workflows

NetEase (Menghuan Studio) — *Senior Engine Programmer*

Oct 2020 – Feb 2021

- Runtime virtual texture pre-development.

IGG (SkyUnion Studio, Fuzhou) — *Senior Graphics Programmer*

Sep 2018 – Oct 2020

- Replaced Unity's default shadow system with a custom shadowmap pipeline
- Created an instancing-based vegetation rendering system
- Implemented large-scale terrain streaming using Virtual Texture
- Developed an infinite water simulation based on Projected Grid
- Built rendering systems for a startup project's early prototype

Shenzhen Tuoshang Digital — *Technical Partner*

Mar 2017 – Jul 2018

- Physical base rendering (PBR) for realistic car materials
- Strategic decision of rendering solutions for the game.
- Car physics development.
- Game tools development.

Tencent (QQSpeed Project) — *Senior Engineer (L10)*

Feb 2012 – Mar 2017

- Develop, optimize, and maintain the Gamebryo engine in the QQSpeed project.
- Global illumination system development associated with AutoDesk Beast for the QQSpeed project.
- Features of the gameplay system for the project DZS.

Langqi Network Technology — *Senior Game Engine Programmer*

Jun 2011 – Feb 2012

- Led engine development for MMORPG project “XiYou 3” .

Youyi Digital Technology — *Game Engine Programmer*

Jul 2007 – May 2011

- Developed 3D in-house engine (DX9/DX11/OpenGL) and core rendering modules
- Built UI batching, font systems, and gameplay features

VTRON Technologies — *Software Engineer*

Jul 2004 – Jun 2007

- Vtron Media Design Platform (based on DirectX 9).
- IDB Digital Board, Newspaper Reading Software.

Education

South China Agricultural University

Bachelor of Computer Science | Sep 2000 – Jul 2004

Language

- **Chinese:** Native
- **English:** Fluent. IELTS Academic overall 6.5(2025.11)

Personal projects

- **unity-raytracing** – A ray tracing demo built with Unity’ s compute shader and ray tracing shader API
- **SRPRendering** – A modular rendering pipeline built on Unity’ s Scriptable Render Pipeline and RenderGraph